

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

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STEP1: CREATE TABLE

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Server Enterprise Explorer' tree shows the 'Databases' folder expanded, with 'bank_transactions' selected. The 'Properties' pane on the right shows the 'Table Properties' for 'bank_transactions'. The 'Columns' tab is active, showing the following columns:

Column Name	Column Type	Column Length	Column Scale	Column Nullability
txn_id	INT			PRIMARY KEY
customer_name	VARCHAR	50		
branch_name	VARCHAR	50		
transaction_type	VARCHAR	20		
amount	DECIMAL	10	2	
transaction_date	DATE			

The 'Output' pane at the bottom shows the execution of the SQL script. The 'Action Output' tab is selected, displaying the following message:

#	Time	Action	Message
23	20:31:32	CREATE TABLE bank_transactions (txn_id INT P...	0 row(s) affected

STEP 2: INSERT DATA

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cu

```

121 • INSERT INTO bank_transactions VALUES
122     (101,'VAISHNAVI','Hyderabad','Deposit',5000,'2024-01-05'),
123     (102,'LIKHITHA','Hyderabad','Withdrawal',2000,'2024-01-06'),
124     (103,'SUSHMA','Vijayawada','Deposit',12000,'2024-01-08'),
125     (104,'AKHILA','Vizag','Deposit',8000,'2024-01-10'),
126     (105,'DIVYA','Hyderabad','Withdrawal',3500,'2024-01-11'),
127     (106,'DISKHITHA','Vizag','Deposit',15000,'2024-01-12'),
128     (107,'ARCHANA','Vijayawada','Withdrawal',1000,'2024-01-13'),
129     (108,'BABY','Hyderabad','Deposit',9000,'2024-01-14'),
130     (109,'SAI','Vizag','Withdrawal',4000,'2024-01-15'),
131     (110,'KUSHAL','Vijayawada','Deposit',11000,'2024-01-16');
132

```

Conti

Output

Action Output ▼

#	Time	Action	Message
❌ 24	20:31:35	CREATE TABLE bank_transactions (txn_id INT P...	Error Code: 1050. Table 'bank_transactions'
✅ 25	20:33:37	INSERT INTO bank_transactions VALUES (101,VAI...	10 row(s) affected Records: 10 Duplicates:

STEP 3: SUM ()

```

132 • SELECT SUM(amount) AS Total_Amount
133 FROM bank_transactions;
134
135

```

Result Grid |
|

 Filter Rows:
|
 Export:
|
 V

Total_Amount
70500.00

Step 4: AVG ()

```

135 • SELECT AVG(amount) AS Average_Transaction
136 FROM bank_transactions;
137
138

```

Result Grid |
|

 Filter Rows:
|
 Export:
|
 Wrap Cell Co

Average_Transaction
7050.000000

STEP 5: MAX ()

```
.37 • SELECT MAX(amount) AS Highest_Transaction
.38 FROM bank_transactions;
.39
```

Result Grid	Filter Rows:	Export:
Highest_Transaction		
15000.00		

STEP : MIN ()

```
ariat 140 • SELECT MIN(amount) AS Lowest_Transaction
141 FROM bank_transactions;
142
143
```

Result Grid	Filter Rows:	Export:	Wrap Cell C
Lowest_Transaction			
▶ 1000.00			

STEP 7 : COUNT()

```
ariat 143 • SELECT COUNT(*) AS Total_Transactions
144 FROM bank_transactions;
145
```

Result Grid	Filter Rows:	Export:	V
Total_Transactions			
▶ 10			

STEP 8 : WHERE + SUM()

```
145 • SELECT SUM(amount) AS Total_Deposit
146 FROM bank_transactions
147 WHERE transaction_type='Deposit';
148
```

Result Grid

Total_Deposit
60000.00

STEP 9 : GROUP BY

```
148 • SELECT branch_name,
149          SUM(amount) AS Total_Amount
150 FROM bank_transactions
151 GROUP BY branch_name;
152
```

Result Grid

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

STEP 10 : GROUP BY + HAVING

```
153 • SELECT branch_name,
154          SUM(amount) AS Total_Amount
155 FROM bank_transactions
156 GROUP BY branch_name
157 HAVING SUM(amount) > 20000;
158
```

Result Grid

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

STEP 11: GROUP BY + ORDER BY

```

159 • SELECT branch_name,
160         SUM(amount) AS Total_Amount
161 FROM bank_transactions
162 GROUP BY branch_name
163 ORDER BY Total_Amount DESC;
164

```

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

STEP 12 : GROUP BY + HAVING + ORDER BY

```

165 • SELECT branch_name,
166         COUNT(*) AS Total_Transactions
167 FROM bank_transactions
168 GROUP BY branch_name
169 HAVING COUNT(*) >= 3
170 ORDER BY Total_Transactions DESC;

```

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

STEP 13: WHERE + COUNT ()

```

171 • SELECT COUNT(*) AS Withdrawals
172 FROM bank_transactions
173 WHERE transaction_type='Withdrawal';

```

Withdrawals
4

STEP 14: WHERE + AVG ()

```

174 • SELECT AVG(amount) AS Avg_Deposit
175 FROM bank_transactions
176 WHERE transaction_type='Deposit';

```

Result Grid		Filter Rows:	Export:
	Avg_Deposit		
▶	10000.000000		

STEP 15: GROUP BY + MAX()

```

178 • SELECT branch_name,
179           MAX(amount) AS Highest_Amount
180 FROM bank_transactions
181 GROUP BY branch_name;

```

Result Grid		Filter Rows:	Export:
	branch_name	Highest_Amount	
▶	Hyderabad	9000.00	
	Vijayawada	12000.00	
	Vizag	15000.00	

STEP 1: GROUP BY + MIN ()

```

183 • SELECT branch_name,
184           MIN(amount) AS Lowest_Amount
185 FROM bank_transactions
186 GROUP BY branch_name;

```

Result Grid		Filter Rows:	Export:
	branch_name	Lowest_Amount	
	Hyderabad	2000.00	
	Vijayawada	1000.00	
	Vizag	4000.00	

Step 17: WHERE + ORDER BY


```
187 • SELECT *
188 FROM bank_transactions
189 WHERE amount > 8000
190 ORDER BY amount DESC;
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	DISKHITHA	Vizag	Deposit	5000.00	2024-01-12
	103	SUSHMA	Vijayawada	Deposit	12000.00	2024-01-08
	110	KUSHAL	Vijayawada	Deposit	11000.00	2024-01-16
	108	BABY	Hyderabad	Deposit	9000.00	2024-01-14
✱	NULL	NULL	NULL	NULL	NULL	NULL

Step 18 : GROUP BY + COUNT ()

```
191 • SELECT branch_name,
192         COUNT(*) AS Customer_Count
193 FROM bank_transactions
194 GROUP BY branch_name;
```

Result Grid





Filter Rows:

Export

	branch_name	Customer_Count
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

STEP 19 : HAVING + COUNT()

```
195 • SELECT branch_name,
196         COUNT(*) AS Total_Transactions
197 FROM bank_transactions
198 GROUP BY branch_name
199 HAVING COUNT(*) > 3;
```

Result Grid


Filter Rows:

Export: 

	branch_name	Total_Transactions
▶	Hyderabad	4

STEP 20: GROUP BY + ANG ()

```

199     HAVING COUNT(*) > 3;
200 •   SELECT branch_name,
201         AVG(amount) AS Avg_Amount
202     FROM bank_transactions
203     GROUP BY branch_name;

```

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

STEP 21: WHERE + GROUP BY + HAVING

```

204 •   SELECT branch_name,
205         SUM(amount) AS Total_Deposit
206     FROM bank_transactions
207     WHERE transaction_type='Deposit'
208     GROUP BY branch_name
209     HAVING SUM(amount) > 15000;

```

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

STEP 22: GROUP BY + HAVING + ORDER BY

```

210 •   SELECT branch_name,
211         AVG(amount) AS Avg_Amount
212     FROM bank_transactions
213     GROUP BY branch_name
214     HAVING AVG(amount) > 7000
215     ORDER BY Avg_Amount DESC;

```

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

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